Credit line increase: what to ship and what to test next

Data: 1,000,000 members (500,226 control / 499,774 treatment), joined Jun 1 – Aug 9, snapshot Aug 25. Retention = account still open N days after joining, among members with at least N days of tenure.

Decision: ship to members with room, switch it off for members with none. Not a plain ship. The average result is flat because it adds a large gain and a large loss. Members whose ceiling is above their limit get the flow, with the slider capped at their room. Members with no room see no entry point. Projected 30-day retention: **~92.4% vs 87.8%** in control (+4.6 pts). A 5% no-feature holdback among members with room checks that the gain holds for the revised flow (see allocation).

EVIDENCE

30-DAY RETENTION	SHARE	TREATMENT	CONTROL	DIFFERENCE
Everyone	100%	87.8%	87.8%	0.0 pts
Room under ceiling (room > 0)	77%	93.8%	87.8%	**+6.0 ± 0.2**
No room (room = 0)	23%	68.0%	87.9%	**–19.9 ± 0.4**

- **Every request from a no-room member was declined: 110,199 of 110,199.** The app let them ask for something that could not be approved, and ran a soft inquiry to tell them so.
- **The decline does the damage, not the reason or the ask.** Declined members keep ~68% whatever the reason code (67.8–68.6%) and whatever they asked for (67.6–68.3%). No-room treatment members close at 3.5%/day in their first week vs 0.43%/day in control.
- **The money doesn't drive the gain.** Approved members retain 93.6–94.2% at every ask size and every share of room taken. Members with room who never asked retain 93.5%.
- **The gain arrives late and for everyone in treatment.** Among members still open at day 15, 0.2–0.3% close by day 30 in every treatment subgroup (approved, never asked, even declined) vs 6.55% in control. The room > 0 gain grows to +7.5 pts at 60 days.
- **It's stable:** same sign and size in all 8 readable join weeks and all four starting limits; room and limit mix are balanced across groups. The 14-day ITT reads –6.0 pts only because the decline burst comes first.
- Data notes: 18,438 approved asks are for $50, below the $100 API minimum (same retention). Approved members show a closure spike the day after asking, but it's timing, not harm: 5.98% closed by day 14 vs 6.00% in control.

ALLOCATION: TWO SEPARATE SPLITS, BY ROOM AT SIGNUP

MEMBERS	SHARE	SPLIT
Room > 0	77%	95% revised request flow · 5% holdback with no increase feature
Room = 0	23%	50% A · hidden · 50% B · limit status card (the next experiment, below)

The holdback confirms the gain persists now that the flow has changed. It is not part of the no-room experiment, and 5% is an operational choice, not an allocation sized from this analysis.

NEXT EXPERIMENT: WHAT NO-ROOM MEMBERS SEE (23% OF SIGNUPS)

A · Hidden. No entry point, the same Home the control group saw. Expected ~87.9%.

B · Limit status card. No ask, no soft inquiry. A card on Home states the current limit, that no increase is available, and the specific reasons. Nothing else (no advice), so the test isolates the explanation; no promise of a future increase.

Why this one. Treatment members who never asked, and even declined members who stayed past day 14, almost stopped closing. Something about the experience helps beyond the money. B tests whether no-room members can get that without the decline. If B reaches the room > 0 level, that's +5.9 pts on 23% of members ≈ **+1.4 pts overall**, the largest lever left.

Primary metric: 30-day retention. **Guardrail:** 7-day closure rate, checked daily. Stop B if it runs 1 pt above A, since B could feel like a decline without the ask.

Size: a 1-pt effect at an 88% baseline needs ~16,600 members per arm (α 0.05, power 0.8). That's ~23,000 no-room signups a week, so ~1.5 weeks to enrol, then 30 days to read.

Deprioritised: showing members with room their headroom up front (never-askers already retain 93.5%, so there's little to gain); amount defaults (size has no effect); a softer decline screen for no-room members (still a doomed ask).

Open questions: never-askers are a self-selected 5%. Why closures nearly stop after day 14 isn't explained by these two tables. B needs the API to expose the reason codes it would return (added to the mock as `increase_eligibility`).

WHAT CHANGES IN THE APP (BOTH ARMS)

No doomed asks: the entry point shows only when at least $100 fits under the ceiling, and the slider stops at the member's room. **Declines name the reasons:** the baseline said only "Declined"; now each `reason_code` is in plain words, on the result and on Home afterwards. **Soft inquiry explained** before submitting and after a decline. **One request, said up front**, and Home then shows the decision instead of a button that disappears. **Approval is live at once**, and Home reloads the account. **Failures are safe:** after a lost response the app reads the account before any resubmit, and stays "can't confirm" if it can't.